

Pendulum Modeling TF

EGR 345

GVSU

Outline

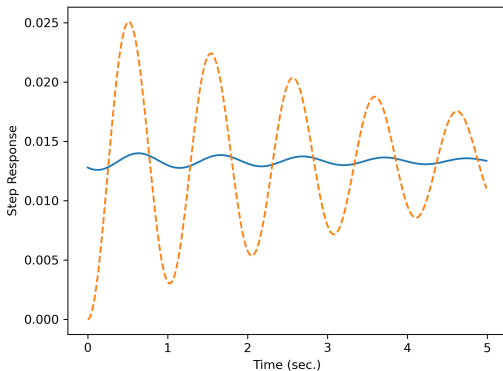
- debugging my MSD example from last lecture
- pendulum modeling
 - displacement input TF

Teaching Philosophy Issue

- I don't script my classes and work out all my examples ahead of time...
 - bug or feature?

Step Response Issue from Last Time

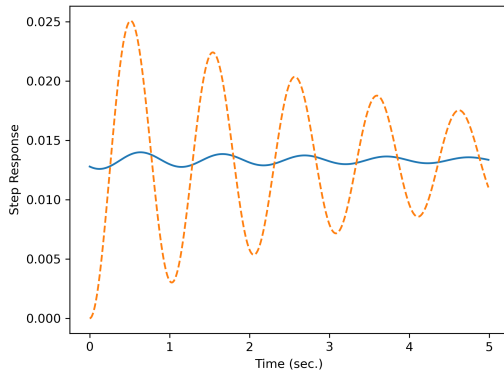
- don't blurt out the answer yet
- I see two issues with this graph, one quantitative and one qualitative



Step Response Issue Continued

Known form:

$$y(t) = C_1 \cdot 1(t) + e^{-at} (C_2 \sin(\omega_d t) + C_3 \cos(\omega_d t))$$

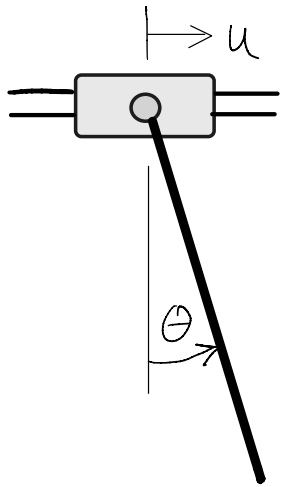


Pendulum Modeling

Is a pendulum basically the same as a mass/spring/damper system?

- what did you learn in last week's lab?
 - initial condition response
 - impulse response from tap/whack
- what about voltage input to the motor on the cart?

Is a pendulum basically the same as a mass/spring/damper system?



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